

Experiment E1: equivalence-scale preferences with sequential births

Design and calibration contract for the `intergen_eqscale_seq` fork

Fertility–housing project

July 18, 2026 — an experiment; production and both prior forks untouched

1. What the experiment asks

Two structural changes at once, each motivated by a measured failure of the production model, recalibrated on the unchanged 15-moment target system so the fit is comparable to M5. First, preferences lose their Stone–Geary intercepts, so no income realization is ever infeasible and the income process can carry literature-level risk ($\sigma_z = 0.20$, set externally) with no safety-net apparatus at all. Second, fertility becomes sequential: households try for a first child, and one-child households may later try for a second, each attempt succeeding with the age-declining fecundity probability of the July-18 design note. The question “take a look” will answer: can this architecture match the same data the production model matches, and what do its timing margins and honest income risk do to the objects the production model misses (the wealth tail, young liquid wealth, the childlessness composition)?

2. Preferences: why not textbook Barten scales

The obvious equivalence-scale implementation — divide each good by a child-dependent scale inside the existing composite — is *inert* here: the period composite is Cobb–Douglas, so good-specific scales enter log-additively and cancel from the allocation between consumption and housing. Children would stop tilting housing demand entirely, removing the paper’s mechanism. The experiment therefore prices children through the two devices that do work under Cobb–Douglas:

$$u(c, h; n, s) = \tilde{e}(n, s) \frac{(c^{\alpha(n, s)} h^{1-\alpha(n, s)})^{1-\sigma}}{1-\sigma} + \psi n_k, \quad \tilde{e}(n, s) = 1 + \gamma_e n_k,$$

where $\tilde{e}$ is the flow-utility multiplier itself (equal to the equivalence scale raised to $\sigma - 1$; at the maintained $\sigma = 2$ the two coincide, so γ_e parameterizes the scale directly), with a *share shifter* $\alpha(n, s) = \text{clip}(\alpha_0 - (\delta_\alpha^{\text{jump}} + \delta_\alpha n_k) \mathbf{1}\{\text{children present}\}, 0.05, 0.95)$ — children tilt expenditure toward housing, the demographic-taste-shifter tradition of the consumption literature — and a *composite deflator* $e(n, s) = 1 + \gamma_e n_k$, the utility cost of children (with $\sigma = 2$ the deflator multiplies the negative CRRA term, so larger families need more of both goods to reach the same standard of living). There are no intercepts: marginal utility diverges as consumption or housing services approach zero, every strictly positive income draw admits a feasible choice, and the empty-budget-set machinery (gates, censoring, floors) has nothing to catch. The honest cost, stated once: the childless necessity structure is gone — poor childless households are no longer disciplined by a committed bundle — so the income and tenure gradients must be carried by the borrowing constraint, prices, and the two new parameters. Whether they can be is part of what the recalibration tests.

3. Fertility: the sequential margin

As specified in the July-18 design note, with the v2 extension now active: childless households choose {wait, try for the first child}; households with one child and the window open choose {stop, try for the second}; attempts succeed with probability π_j (external schedule, illustrative until the demographic fit); a second birth lands in the existing two-children raising state with the *youngest-child clock restarted*. The state grid does not grow: the collapse of per-child ages to one clock — the reason sequencing was originally dropped — is retained, at the approximation that the first child’s maturation rides the youngest’s restarted clock (small against an 18-year expected duration). New measured objects: first- and second-birth attempt and realized hazards by age, the flow progression from one to two children, and the chosen-versus-clock childlessness split. TFR and completed-fertility measurement are unchanged.

4. The E1 calibration contract

Torch, 8 chains $\times$ 3:55 (the M5 template), smoke first. Targets: the 15 M5 moments, weights unchanged — timing moments are reported diagnostics, not targets, until their data counterparts are built. Twelve free parameters, over-identified by construction:

Parameter	Domain (transform)	Note
β (annual)	[0.80, 0.9995] (discount)	external floor 0.94 still respected in reporting
α_0 (alpha_cons)	[0.02, 0.98] (logit)	childless housing share
δ_α^{jump}	[0, 0.25] (softzero)	share tilt, any child
δ_α	[0, 0.25] (softzero)	share tilt per child
γ_e	[0, 3] (softzero)	composite deflator per child
ψ (psi_child)	$[-3, 3]$ (asinh)	child utility
κ_f	[0.02, 50] (log)	fertility logit scale
χ	[0.10, 5] (log)	
H_0	[0.20, 80] (log)	
θ_0	[0, 8] (softzero)	bequest
θ_1	[0.02, 16] (log)	bequest luxury
κ_T (tenure_choice_kappa)	[0, 0.12] (softzero)	

External restrictions (none free, none silent): $\theta_n = 0$; $\sigma = 2$; $\phi = 0.80$; income process $\rho_z = 0.9602$, $\sigma_z = 0.20$ (SS-range, *the experiment’s point*); fecundity $(\omega_1, \omega_2, \bar{a}) = (0.02, 0.134, 45)$ illustrative pending the demographic fit; no transfer floor; the five Stone–Geary floor parameters do not exist in this specification; entry wealth, survival, housing menus, taxes as in M5. Seeds: M5 winner for shared coordinates; $(\delta_\alpha^{jump}, \delta_\alpha, \gamma_e) = (0.10, 0.05, 0.5)$.

Reading protocol for the result: report the full 15-row target-fit table against M5’s; the timing diagnostics; the estate p90/p50 and young liquid wealth (the two objects honest risk moves in opposite directions in the production model); and the childlessness composition. This is an experiment: a failure to fit is itself the answer to a question worth asking, and nothing in production moves either way.